

SmartClinic Management System

Database Management System

Project Report

Project Type: Web-Based Hospital & Clinic Management System

Technologies Used: PHP, MySQL, HTML, CSS, Bootstrap, JavaScript

Database: MySQL Database Management System

Development Environment: XAMPP, VS Code, phpMyAdmin

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Course: Database Management System

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1. Abstract

The SmartClinic Management System is a web-based healthcare management platform developed to digitize and automate the operations of clinics and hospitals. The system provides separate panels for administrators, doctors, and patients, allowing efficient management of appointments, prescriptions, patient records, billing, and notifications. The project focuses on improving operational efficiency, reducing paperwork, and ensuring secure handling of medical information.

2. Project Objectives

- To create a centralized hospital management platform.
- To automate appointment booking and patient management.
- To maintain digital medical records securely.
- To provide role-based access for Admin, Doctors, and Patients.
- To improve communication using email verification and notifications.
- To generate billing and reports efficiently.

3. Major Features of the System

- Secure Login and Authentication System
- Patient Registration with OTP Verification
- Forgot Password Functionality with Email Recovery
- Doctor Management Module
- Appointment Booking and Scheduling
- Prescription Management
- Medical Records Management
- Billing and Invoice Generation
- Admin Dashboard and Reports
- Patient Notifications and Status Tracking

4. Development Methodology

The project was developed using the Incremental Development Methodology. Each module of the system was developed and tested separately before integrating it into the complete application. This approach allowed the team to identify bugs early and continuously improve the functionality of the system.

4.1 Technologies and Tools

Technology	Usage
Frontend	HTML, CSS, Bootstrap, JavaScript
Backend	PHP
Database	MySQL
IDE	Visual Studio Code

Server	XAMPP Apache Server
Version Control	Git and GitHub

5. System Workflow

1. User accesses the SmartClinic system through the login page.
2. Patients can register using signup functionality with OTP email verification.
3. Verified patients log in and book appointments with doctors.
4. Doctors manage schedules, prescriptions, and patient medical records.
5. Admin monitors all system activities including doctors, patients, appointments, and reports.
6. The billing system generates invoices and maintains payment records.
7. Notifications and updates are delivered to users through the system.

6. Database Design

The database was designed using MySQL and normalized to reduce redundancy and improve data consistency. Different tables were created for users, doctors, patients, appointments, prescriptions, notifications, and billing records. Primary and foreign keys were implemented to maintain relationships between tables.

7. System Modules

Admin Module

- Manage doctors and patients
- Monitor appointments and reports
- Handle billing operations
- View system-wide analytics

Doctor Module

- Manage appointments
- Add prescriptions
- Update patient medical records
- Manage doctor schedule

Patient Module

- Register and verify account
- Book appointments
- View prescriptions and medical records
- Receive notifications

8. Security Features

- Password-protected authentication system
- OTP-based email verification
- Role-based access control
- Protected database connection file using .gitignore
- Session management for secure login handling

9. Challenges Faced During Development

During development, several technical challenges were encountered, including database relationship management, role-based authentication, secure email verification implementation, and synchronization between different modules. These issues were resolved through debugging, testing, and incremental development.

10. Future Enhancements

- Online payment gateway integration
- Video consultation feature
- AI-based appointment recommendations
- Mobile application development
- Cloud database hosting
- Advanced analytics dashboard

11. Team Member Contributions

Although the project was collaboratively submitted, responsibilities were divided between two members for documentation and presentation purposes.

Team Member	Responsibilities
Adnan Hatim	Worked on backend development using PHP and MySQL, designed and managed database relationships, implemented authentication and login systems, developed appointment booking functionality, integrated OTP verification and forgot password modules, and performed debugging and testing.
Murtaza Hunaid	Worked on frontend design using HTML, CSS, Bootstrap, and JavaScript, developed user interface components, assisted in module integration, managed documentation and project reports, performed system testing, and helped with GitHub version control and presentation preparation.

12. Conclusion

The SmartClinic Management System successfully demonstrates the practical implementation of database management concepts in a real-world healthcare environment. The project improves efficiency, security, and accessibility of hospital operations through automation and digital management techniques.